



Computer System and Architecture

Machine Problem

MIDTERM

RAYMOND RICOHERMOSO JR.	HADJI TEJUCO
Name of Student	Name of Professor
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Data Performed	Date Submitted



Paste the source code and explain the functionalities and take a screenshot of the final output.

```

1. Name: Ram
2. Email: rcricohermosojr@fit.edu.ph
3. Blog/Website: FEU Tech
4. First big achievement: Gold Medalist
5. First risk taken: Not reviewing for a test
6. First time happy: My first birthday
7. Favorite Color(s): Red
8. Favorite Perfume: None
9. Favorite Music genre: Rock
10. Favorite Singer(s): Radiohead
11. Favorite Song: Creep
12. Favorite Food: Buldak
13. Weekend activity: Sleeping
14. Hobbies: Music and Games
15. Favorite TV Show: Game of Thrones
16. Favorite Movie: Spider-Man: No Way Home
17. Favorite Book: ASOIAF Series
18. Favorite Celebs: Tom Holland
19. Role Model: Peter Parker
20. Ambition: To become successful
21. Personal Motto: Be Better

```

```

 @->->--                               MY SLAMBOOK                                <3 <3 <3

[ ABOUT ME ]                             +------+
Name      : Ram                          | PHOTO   |
Email     : rcricohermosojr@fit.edu.ph    | HERE    |
Blog/Website: FEU Tech                    +------+

First big achievement       : Gold Medalist
First risk I ever took     : Not reviewing for a test * : . * . : *
First time completely happy: My first birthday

[ MY FAVES ]
Color/s : Red                Singer/s: Radiohead
Perfume  : None              Song    : Creep
Music    : Rock               Food    : Buldak

[ HOBBIES ]
Weekend activity: Sleeping
Hobbies         : Music and Games
TV Show        : Game of Thrones
Movie          : Spider-Man: No Way Home
Book           : ASOIAF Series
Celebs         : Tom Holland
Role model     : Peter Parker

[ AMBITION ]
Ambition       : To become successful

[ MOTTO ]
Be Better

                                     />             <\
                                    //              \\\
</ //   ____   \\ \>
\\ \\\ (oo) // //
'----X{II}X----'
      //=/\\\/\\=\\
-----/| = = |\\/\\----'
// \\\ =\\\/= / \\\\
(( '~---' ))
\\                      / hJw

```



SOURCE CODE:

```
; MIDTERM EXAM PROJECT IN COMPUTER SCIENCE AND ARCHITECTURE  
; Assembly  
; RICOHERMOSO, RAYMOND JR.
```

```
section .data
```

```
CLEAR_SCR db 27, "[2J", 27, "[1;1H", 0  
NEWLINE db 10, 0
```

```
; Interactive Input Prompts
```

```
PROMPT_TITLE db "=== FILL OUT YOUR SLAMBOOK ===", 10, 10, 0
```

```
P_NAME db "1. Name: ", 0  
P_EMAIL db "2. Email: ", 0  
P_BLOG db "3. Blog/Website: ", 0  
P_ACHIEV db "4. First big achievement: ", 0  
P_RISK db "5. First risk taken: ", 0  
P_HAPPY db "6. First time happy: ", 0  
P_COLOR db "7. Favorite Color(s): ", 0  
P_PERF db "8. Favorite Perfume: ", 0  
P_MUSIC db "9. Favorite Music genre: ", 0  
P_SING db "10. Favorite Singer(s): ", 0  
P_SONG db "11. Favorite Song: ", 0  
P_FOOD db "12. Favorite Food: ", 0  
P_WEEKEND db "13. Weekend activity: ", 0  
P_HOBBIES db "14. Hobbies: ", 0  
P_TV db "15. Favorite TV Show: ", 0  
P_MOVIE db "16. Favorite Movie: ", 0  
P_BOOK db "17. Favorite Book: ", 0  
P_CELEBS db "18. Favorite Celebs: ", 0  
P_ROLE db "19. Role Model: ", 0  
P_AMB db "20. Ambition: ", 0  
P_MOTTO db "21. Personal Motto: ", 0
```

```
; Section Labels
```

```
OUT_HEADER db "MY SLAMBOOK", 0  
HDR_ABOUT db "[ ABOUT ME ]", 0  
LBL_NAME db "Name : ", 0  
LBL_EMAIL db "Email : ", 0  
LBL_BLOG db "Blog/Website: ", 0
```



```
LBL_ACHIEV db "First big achievement : ", 0
LBL_RISK db "First risk I ever took : ", 0
LBL_HAPPY db "First time completely happy: ", 0
```

```
HDR_FAVES db "[ MY FAVES ]", 0
LBL_COLOR db "Color/s : ", 0
LBL_PERF db "Perfume : ", 0
LBL_MUSIC db "Music : ", 0
LBL_SING db "Singer/s: ", 0
LBL_SONG db "Song : ", 0
LBL_FOOD db "Food : ", 0
```

```
HDR_HOBBY db "[ HOBBIES ]", 0
LBL_WEEKEND db "Weekend activity: ", 0
LBL_HOBBIES db "Hobbies : ", 0
LBL_TV db "TV Show : ", 0
LBL_MOVIE db "Movie : ", 0
LBL_BOOK db "Book : ", 0
LBL_CELEBS db "Celebs : ", 0
LBL_ROLE db "Role model : ", 0
```

```
HDR_AMB db "[ AMBITION ]", 0
LBL_AMB db "Ambition : ", 0
```

```
HDR_MOTTO db "[ MOTTO ]", 0
```

```
; --- ASCII ART ELEMENTS ---
```

```
; Outer Frame Border Rows
```

```
BORDER_TOP db "+-----"
+", 0
BORDER_MID db "| "
BORDER_BOT db "+-----"
+", 0
```

```
; Photo Frame
```

```
ART_PHOTO1 db "+-----+", 0
ART_PHOTO2 db "| PHOTO |", 0
ART_PHOTO3 db "| HERE |", 0
ART_PHOTO4 db "+-----+", 0
```

```
; Decorative Art Doodles
```

```
ART_FLOWER db "@->->--", 0
```



```
ART_HEARTS db "<3 <3 <3", 0
ART_STARS db "* : . * : . *", 0
```

```
; Spider ASCII Art (11 lines)
SPIDER_1 db "  />  <\", 0
SPIDER_2 db "  //   \\", 0
SPIDER_3 db " </ //   \\ >", 0
SPIDER_4 db " \\ \\ (oo) // //", 0
SPIDER_5 db " '---X{II}X.---", 0
SPIDER_6 db "  //=/\\\\=\\", 0
SPIDER_7 db "----/| = = |\\----", 0
SPIDER_8 db "  //\\ =\\V= / \\\\", 0
SPIDER_9 db " (( `-.-' ))", 0
SPIDER_10 db "  \\      /  hjw", 0
```

section .bss

```
; Input Storage Buffers
buf_name resb 64
buf_email resb 64
buf_blog resb 64
buf_achiev resb 64
buf_risk resb 64
buf_happy resb 64
buf_color resb 64
buf_perf resb 64
buf_music resb 64
buf_sing resb 64
buf_song resb 64
buf_food resb 64
buf_weekend resb 64
buf_hobbies resb 64
buf_tv resb 64
buf_movie resb 64
buf_book resb 64
buf_celebs resb 64
buf_role resb 64
buf_amb resb 64
buf_motto resb 64
```

```
pos_buf resb 12
```

section .text

```
global _start
```

MIDTERM EXAM



```
_start:
; =====
; STEP 1: COLLECT USER INPUTS
; =====
push CLEAR_SCR
call print_str

push PROMPT_TITLE
call print_str

push P_NAME
push buf_name
call get_input

push P_EMAIL
push buf_email
call get_input

push P_BLOG
push buf_blog
call get_input

push P_ACHIEV
push buf_achiev
call get_input

push P_RISK
push buf_risk
call get_input

push P_HAPPY
push buf_happy
call get_input

push P_COLOR
push buf_color
call get_input

push P_PERF
push buf_perf
call get_input
```



push P_MUSIC
push buf_music
call get_input

push P_SING
push buf_sing
call get_input

push P_SONG
push buf_song
call get_input

push P_FOOD
push buf_food
call get_input

push P_WEEKEND
push buf_weekend
call get_input

push P_HOBBIES
push buf_hobbies
call get_input

push P_TV
push buf_tv
call get_input

push P_MOVIE
push buf_movie
call get_input

push P_BOOK
push buf_book
call get_input

push P_CELEBS
push buf_celebs
call get_input

push P_ROLE
push buf_role
call get_input



```
push P_AMB
push buf_amb
call get_input
```

```
push P_MOTTO
push buf_motto
call get_input
```

```
; =====
; STEP 2: DRAW PAPER FRAME & COMPLETED SLAMBOOK
; =====
```

```
push CLEAR_SCR
call print_str
```

```
; Draw Top Border
push 1
push 1
call goto_xy
push BORDER_TOP
call print_str
```

```
; Draw Side Borders (Rows 2 to 34)
mov ecx, 2
.draw_frame_sides:
push ecx
push 1
push ecx
call goto_xy
push BORDER_MID
call print_str
pop ecx
inc ecx
cmp ecx, 35
jne .draw_frame_sides
```

```
; Draw Bottom Border
push 1
push 35
call goto_xy
push BORDER_BOT
call print_str
```



; --- Title Header & Top Art ---

push 34

push 2

call goto_xy

push OUT_HEADER

call print_str

push 5

push 2

call goto_xy

push ART_FLOWER

call print_str

push 65

push 2

call goto_xy

push ART_HEARTS

call print_str

; --- About Me & Photo Box ---

push 4

push 4

call goto_xy

push HDR_ABOUT

call print_str

push 6

push 5

call goto_xy

push LBL_NAME

call print_str

push buf_name

call print_str

push 6

push 6

call goto_xy

push LBL_EMAIL

call print_str

push buf_email

call print_str

push 6



```
push 7
call goto_xy
push LBL_BLOG
call print_str
push buf_blog
call print_str

; Photo Frame
push 62
push 4
call goto_xy
push ART_PHOTO1
call print_str
push 62
push 5
call goto_xy
push ART_PHOTO2
call print_str
push 62
push 6
call goto_xy
push ART_PHOTO3
call print_str
push 62
push 7
call goto_xy
push ART_PHOTO4
call print_str
```

```
; --- My Firsts ---
push 6
push 9
call goto_xy
push LBL_ACHIEV
call print_str
push buf_achiev
call print_str
```

```
push 6
push 10
call goto_xy
push LBL_RISK
call print_str
```



```
push buf_risk  
call print_str
```

```
push 6  
push 11  
call goto_xy  
push LBL_HAPPY  
call print_str  
push buf_happy  
call print_str
```

```
; Decorative Stars  
push 62  
push 10  
call goto_xy  
push ART_STARS  
call print_str
```

```
; --- My Faves ---  
push 4  
push 13  
call goto_xy  
push HDR_FAVES  
call print_str
```

```
push 6  
push 14  
call goto_xy  
push LBL_COLOR  
call print_str  
push buf_color  
call print_str
```

```
push 6  
push 15  
call goto_xy  
push LBL_PERF  
call print_str  
push buf_perf  
call print_str
```

```
push 6  
push 16
```



```
call goto_xy  
push LBL_MUSIC  
call print_str  
push buf_music  
call print_str
```

```
push 40  
push 14  
call goto_xy  
push LBL_SING  
call print_str  
push buf_sing  
call print_str
```

```
push 40  
push 15  
call goto_xy  
push LBL_SONG  
call print_str  
push buf_song  
call print_str
```

```
push 40  
push 17  
call goto_xy  
push LBL_FOOD  
call print_str  
push buf_food  
call print_str
```

```
; --- Hobbies ---  
push 4  
push 18  
call goto_xy  
push HDR_HOBBY  
call print_str
```

```
push 6  
push 19  
call goto_xy  
push LBL_WEEKEND  
call print_str  
push buf_weekend
```



call print_str

push 6
push 20
call goto_xy
push LBL_HOBBIES
call print_str
push buf_hobbies
call print_str

push 6
push 21
call goto_xy
push LBL_TV
call print_str
push buf_tv
call print_str

push 6
push 22
call goto_xy
push LBL_MOVIE
call print_str
push buf_movie
call print_str

push 6
push 23
call goto_xy
push LBL_BOOK
call print_str
push buf_book
call print_str

push 6
push 24
call goto_xy
push LBL_CELEBS
call print_str
push buf_celebs
call print_str

push 6



```
push 25
call goto_xy
push LBL_ROLE
call print_str
push buf_role
call print_str
```

```
; --- Ambition ---
push 4
push 27
call goto_xy
push HDR_AMB
call print_str
```

```
push 6
push 28
call goto_xy
push LBL_AMB
call print_str
push buf_amb
call print_str
```

```
; --- Motto ---
push 4
push 30
call goto_xy
push HDR_MOTTO
call print_str
```

```
push 6
push 31
call goto_xy
push buf_motto
call print_str
```

```
; --- DRAW SPIDER ASCII ART (Bottom Right) ---
push 53
push 23
call goto_xy
push SPIDER_1
call print_str
```

```
push 53
```



push 24
call goto_xy
push SPIDER_2
call print_str

push 53
push 25
call goto_xy
push SPIDER_3
call print_str

push 53
push 26
call goto_xy
push SPIDER_4
call print_str

push 53
push 27
call goto_xy
push SPIDER_5
call print_str

push 53
push 28
call goto_xy
push SPIDER_6
call print_str

push 53
push 29
call goto_xy
push SPIDER_7
call print_str

push 53
push 30
call goto_xy
push SPIDER_8
call print_str

push 53
push 31



```
call goto_xy
push SPIDER_9
call print_str
```

```
push 53
push 32
call goto_xy
push SPIDER_10
call print_str
```

```
; Position Cursor outside the frame before exiting
push 1
push 36
call goto_xy
```

```
; Exit Program
mov eax, 1      ; sys_exit
xor ebx, ebx    ; exit code 0
int 0x80
```

```
; =====
; HELPER ROUTINES
; =====
```

```
get_input:
push ebp
mov ebp, esp
```

```
push dword [ebp + 12]
call print_str
```

```
mov eax, 3      ; sys_read
mov ebx, 0      ; stdin
mov ecx, [ebp + 8] ; buffer address
mov edx, 63     ; max length
int 0x80
```

```
mov edi, [ebp + 8]
add edi, eax
dec edi
cmp byte [edi], 10
jne .skip_strip
mov byte [edi], 0
```



.skip_strip:

```
mov esp, ebp
pop ebp
ret 8
```

goto_xy:

```
push ebp
mov ebp, esp
```

```
mov edi, pos_buf
mov byte [edi], 27
inc edi
mov byte [edi], '['
inc edi
```

```
mov eax, [ebp + 8]
call itoa
```

```
mov byte [edi], ';'
inc edi
```

```
mov eax, [ebp + 12]
call itoa
```

```
mov byte [edi], 'H'
inc edi
```

```
mov edx, edi
sub edx, pos_buf
mov eax, 4
mov ebx, 1
mov ecx, pos_buf
int 0x80
```

```
mov esp, ebp
pop ebp
ret 8
```

print_str:

```
push ebp
mov ebp, esp
push edi
```



```
    mov edi, [ebp + 8]
    xor ecx, ecx
.count_loop:
    cmp byte [edi + ecx], 0
    je .print_now
    inc ecx
    jmp .count_loop
```

```
.print_now:
    mov edx, ecx
    mov ecx, [ebp + 8]
    mov eax, 4
    mov ebx, 1
    int 0x80
```

```
    pop edi
    mov esp, ebp
    pop ebp
    ret 4
```

```
itoa:
    push ebx
    push ecx
    push edx
```

```
    mov ebx, 10
    xor ecx, ecx
```

```
.div_loop:
    xor edx, edx
    div ebx
    add edx, '0'
    push edx
    inc ecx
    test eax, eax
    jnz .div_loop
```

```
.pop_loop:
    pop edx
    mov [edi], dl
    inc edi
    loop .pop_loop
```



```
pop edx  
pop ecx  
pop ebx  
ret
```